Cross-Domain NLP Project

Depression Detection → Suicidal Ideation

Technical Summary

Training Dataset	**MDDL — Twitter Depression Detection**
Evaluation Dataset	twitter-suicidal_data — Suicidal Ideation
Task	Binary Text Classification (Cross-Domain Transfer)
Platform	Twitter → Twitter (Topical Domain Shift)
Experiments	TF-IDF / RoBERTa / MentalBERT
Classifiers	6 per experiment (NB, LR, SVM, RF, DT, KNN)

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1. Project Overview

This notebook investigates whether machine learning classifiers trained on depression-related tweets can detect suicidal ideation in a completely separate dataset , without ever seeing suicidal content during training. This is called cross-domain transfer: training on one topic and evaluating on a related but distinct topic.

The core research question is: do depression and suicidal ideation share enough linguistic patterns that a model trained on one can generalise to the other?

Three full experiments were designed and run to answer this, each using a different method of representing text as numbers that a classifier can learn from:

Experiment	Feature Method	Key Idea
A	**TF-IDF**	Word frequency counting: no semantic meaning
B	**RoBERTa**	General-purpose transformer: captures meaning
C	**MentalBERT**	Mental-health-domain transformer: specialised

2. Datasets

2.1 Training Dataset — MDDL

- Source: Twitter API
- Content: Tweets from users who self-reported a depression diagnosis
- Labels: Depressed (1) vs Non-Depressed (0)
- Structure: JSON files organised by user, containing tweets from one month before and after an anchor tweet where the user disclosed their diagnosis
- Important: The anchor tweets themselves were removed to prevent the model from learning trivial signals such as 'I was just diagnosed with depression'

2.2 Evaluation Dataset — twitter-suicidal_data

- Source: Twitter
- Content: Tweets labelled as expressing suicidal ideation or not
- Labels: Suicidal (1) vs Non-Suicidal (0)
- Critical rule: This dataset was NEVER used for training in any experiment: only for evaluating cross-domain transfer

3. Preprocessing

Both datasets were put through the same cleaning pipeline. The steps are listed in the order they were applied:

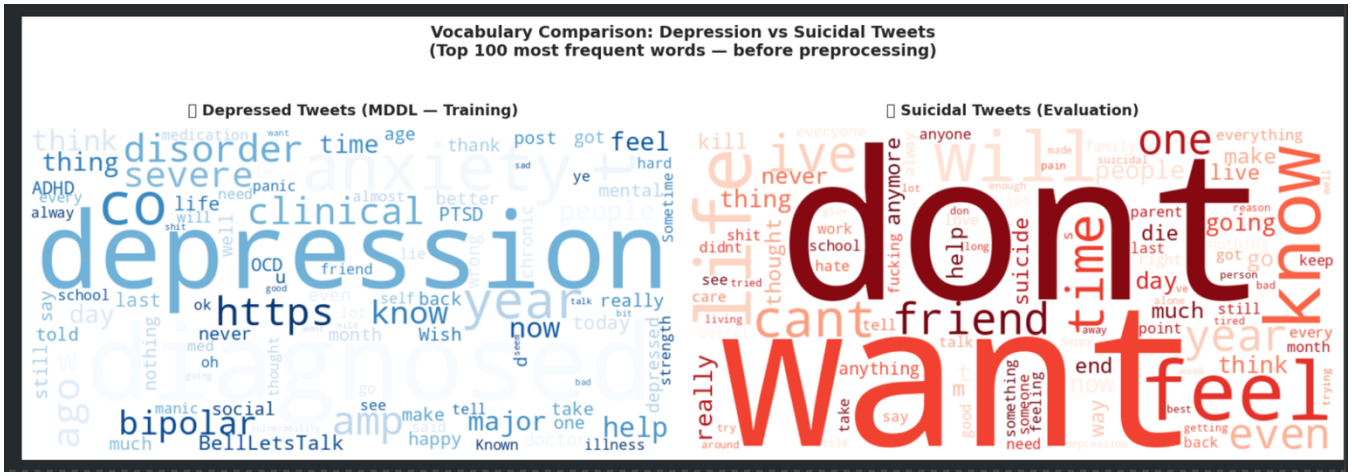
#	Step	Why
1	**Language detection**	Kept only English tweets, langdetect with fixed seed for reproducibility
2	**URL removal**	Hyperlinks carry no semantic meaning for classification
3	**Mention removal**	@username tags identify people, not topic content
4	**Hashtag cleaning**	Stripped # symbol but kept the word itself
5	**Lowercasing**	Standardises text so 'Hopeless' and 'hopeless' are treated the same
6	**Special char removal**	Removed punctuation and non-alphabetic symbols
7	**Stopword removal**	Removed common words (the, is, a) that don't help classification
8	**Lemmatisation**	Reduced words to root form — running → run, hopeless → hopeless

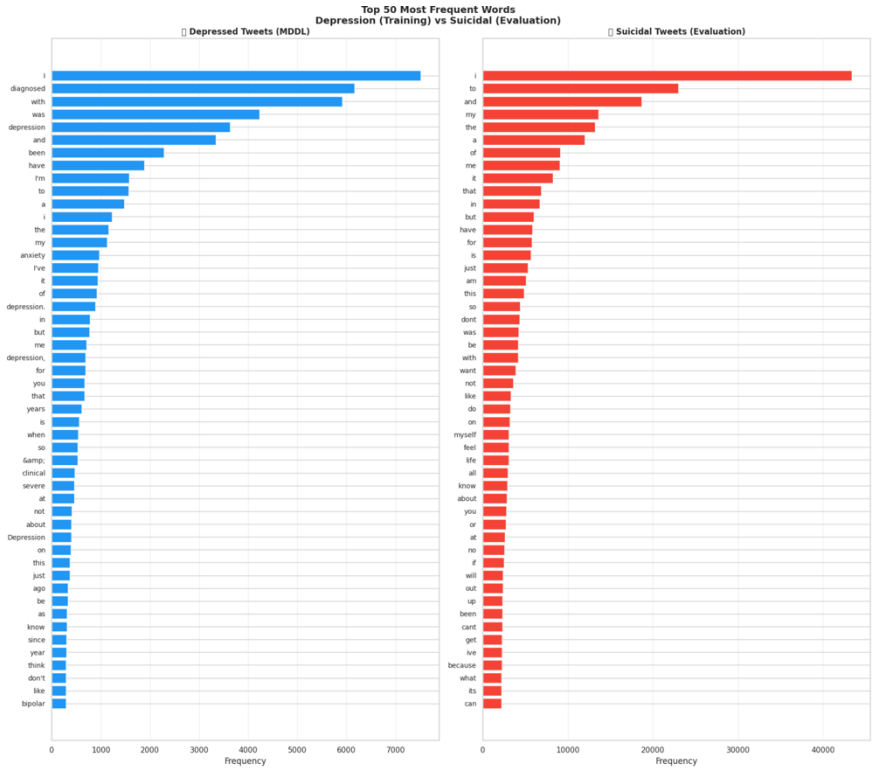
The cleaned datasets were saved to Google Drive after preprocessing so the team never need to be re-processed between sessions.

4. Vocabulary Analysis (EDA)

Before any modelling, we analysed the vocabulary of both datasets to understand how linguistically similar or different they are. Three visualisations were produced:

- Word clouds: visual representation of the most frequent words in each dataset side by side
- Top 50 words bar chart: frequency comparison with exact counts
- Overlap analysis: counted how many of the top 50 words appear in both datasets





4.1 Key Finding — 60% Vocabulary Non-Overlap

Only 20 out of the top 50 words (40%) are shared between the two datasets. The other 60% are unique to each domain:

Shared (20 words)	Depression Only (30 words)	Suicidal Only (30 words)
feel, know, life, like, never, not, nothing, think, time, day, get, help, dont, ive, no, one, people, really, thing, year	depression, depressed, anxiety, mental, disorder, diagnosed, bipolar, ptsd, ocd, adhd, clinical, doctor, chronic, severe, bellletstalk...	want, die, kill, suicide, suicidal, end, cant, hate, live, anyone, anymore, everything, everyone, care, need, going, would, friend...

📌 *This 60% non-overlap is the root cause of all cross-domain failures seen in Experiment A. The depression dataset is dominated by clinical and diagnostic terminology while the suicidal dataset uses action-oriented and interpersonal distress language.*

5. Experiment Design

All three experiments follow the exact same protocol so results are directly comparable. The only difference is the feature representation method.

Component	Detail
Training data	Full cleaned MDDL corpus (all experiments)
Evaluation data	Full cleaned suicidal dataset (all experiments)
Classifiers	Naive Bayes, Logistic Regression, SVM (Linear), Random Forest, Decision Tree, KNN
In-domain split	User-level 80/20 train/test split
Cross-validation	5-fold Stratified CV on training split
Metrics	Accuracy, Precision, Recall, F1, AP Score (Average Precision)
Baseline	Majority-class DummyClassifier as performance floor

5.1 Why User-Level Splitting Matters

A critical methodological decision was splitting the in-domain data at the user level rather than the tweet level. If we split randomly at the tweet level, the same user's tweets can appear in both training and test sets. Since people have consistent writing styles, the model can learn to recognise individual users rather than genuinely generalising, this inflates scores artificially and is called data leakage.

By splitting at the user level, we guarantee that no user's tweets appear in both train and test. This gives honest estimates of how well the model generalises to new users, not just new tweets from users it already knows.

5.2 Why a Dummy Baseline Was Added

A DummyClassifier that always predicts the majority class was included in every cross-domain experiment. Any real model that cannot beat this baseline is not learning anything useful. This baseline also exposes a trap: under class imbalance, a dummy that predicts everything as suicidal achieves F1 of 0.4771 simply because Recall = 1.0. Real models score lower F1 not because they are worse, but because they are more selective. The meaningful comparison is Recall and AP Score, not raw F1 against the dummy.

6. Experiment A — TF-IDF + Traditional Classifiers

6.1 What TF-IDF Does

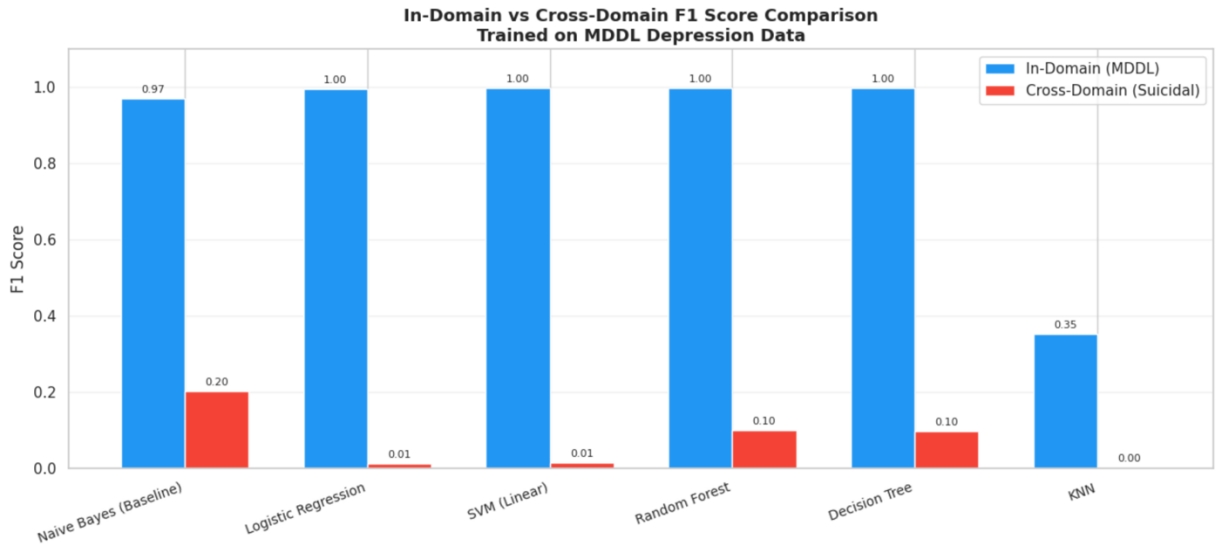
TF-IDF (Term Frequency-Inverse Document Frequency) converts each tweet into a vector of numbers where each number represents how important a specific word is in that tweet relative to the whole dataset. It has no understanding of meaning, it only counts how often words appear. Two separate vectorizers were used: one fitted on the full MDDL corpus for the cross-domain pipeline, and one fitted on only the 80% training split for the in-domain pipeline.

6.2 In-Domain Results

All models except KNN achieved Test F1 above 0.96, confirming that TF-IDF features are highly discriminative within the depression domain:

Model	CV F1 (mean ± std)	Acc	Prec	Rec	F1	Note
Decision Tree	0.9978 ± 0.0013	0.9987	0.9987	0.9987	**0.9987**	🥇 Best
Random Forest	0.9979 ± 0.0012	0.9987	1.0000	0.9974	**0.9987**	🥇 Tied
SVM (Linear)	0.9973 ± 0.0019	0.9981	1.0000	0.9961	**0.9980**	
Logistic Regression	0.9923 ± 0.0010	0.9961	1.0000	0.9922	**0.9961**	
Naive Bayes	0.9683 ± 0.0076	0.9690	0.9557	0.9831	**0.9692**	
KNN	0.3210 ± 0.0167	0.6101	1.0000	0.2135	**0.3519**	⚠️ Fails

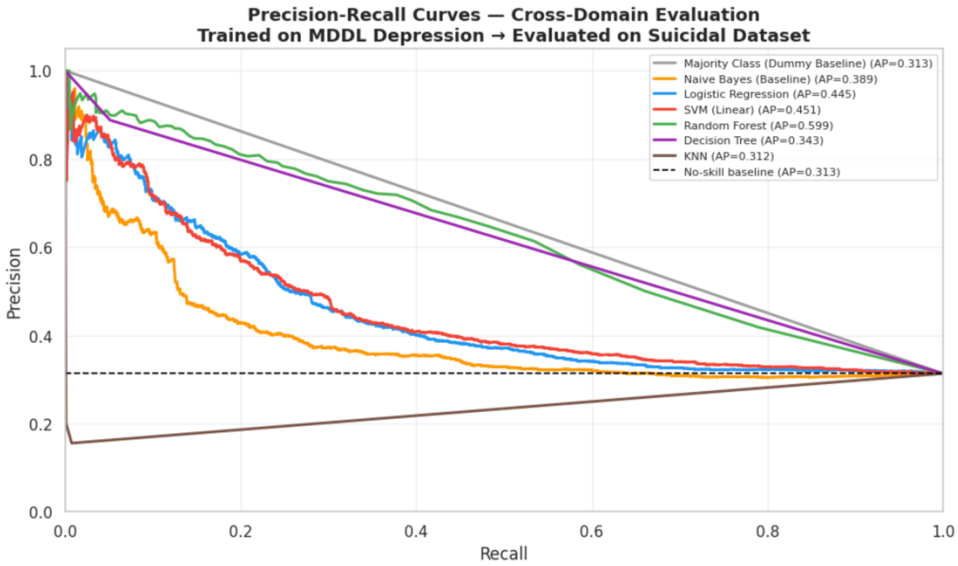
📌 *KNN failed due to the curse of dimensionality, it struggles with high-dimensional sparse TF-IDF vectors where most entries are zero.*



6.3 Cross-Domain Results

Performance collapsed catastrophically for every real classifier when evaluated on the suicidal dataset:

Model	Acc	Prec	Rec	F1	AP	F1 Drop	Note
Dummy Baseline	0.3133	0.3133	1.0000	**0.4771**	0.3133	—	⚠️ Floor
Naive Bayes	0.6900	0.5219	0.1259	**0.2028**	0.3894	**−79.1%**	Best real
Random Forest	0.7014	0.8984	0.0528	**0.0998**	0.5989	**−90.1%**	Best AP
Decision Tree	0.7007	0.8880	0.0510	**0.0964**	0.3426	**−90.4%**	
SVM (Linear)	0.6889	0.9412	0.0073	**0.0146**	0.4514	**−98.5%**	
Logistic Regression	0.6884	0.9286	0.0060	**0.0119**	0.4448	**−98.8%**	
KNN	0.6867	0.5000	0.0005	**0.0009**	0.3123	**−99.7%**	



6.4 Key Findings from Experiment A

- Naive Bayes transferred best: Despite ranking 5th in-domain (F1: 0.9692), it achieved the highest cross-domain F1 (0.2028). It makes soft probabilistic decisions; shared emotional words (nothing, feel, never) still contributed positively in the new domain. SVM and LR learn rigid boundaries that collapse when the feature distribution shifts.
- Random Forest, best AP despite low F1: RF achieved AP Score 0.5989 (nearly double the no-skill baseline) despite F1 of only 0.0998. This means its probability estimates are well-calibrated but its default threshold is too conservative cross-domain. Threshold tuning could make RF the strongest cross-domain TF-IDF model.
- The dominant failure mode is near-zero Recall, not Precision. Models are almost never predicting suicidal at all. SVM Recall = 0.0073 means it correctly identified less than 1% of suicidal tweets.
- KNN collapsed entirely with TF-IDF (F1: 0.0009, Recall: 0.0005) — the worst performance of any model across any condition. This is explained by the curse of dimensionality in high-dimensional sparse feature space.

7. Experiment B — RoBERTa Embeddings

7.1 What RoBERTa Does Differently

Instead of counting words, RoBERTa is a transformer model pre-trained on hundreds of millions of sentences. It converts each tweet into a 768-dimensional vector (called a [CLS] embedding) that captures the meaning of the sentence, not just which words appear.

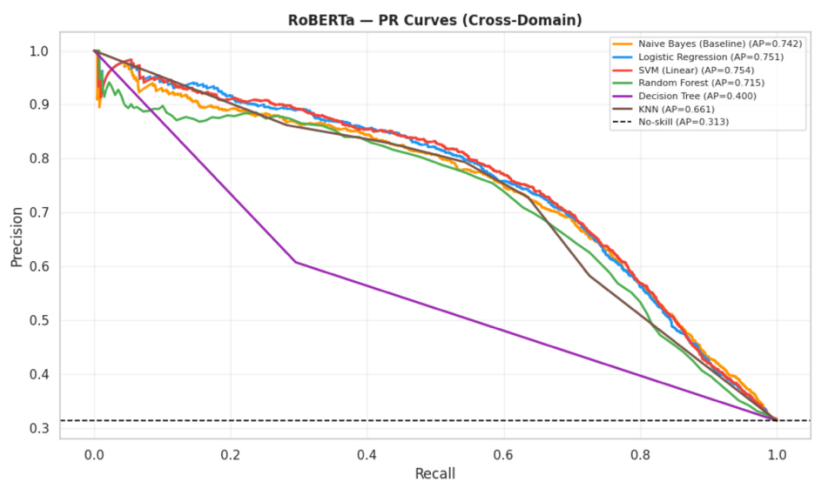
Because RoBERTa was pre-trained on massive text, it already understands that 'I want to end it', 'I feel hopeless', and 'I cannot go on' are semantically similar, even if those phrases never appeared in depression training data. This directly addresses the vocabulary mismatch that destroyed TF-IDF performance.

7.2 Implementation Details

- Model: roberta-base (open access, no restrictions on HuggingFace)
- Embeddings extracted in batches of 32 tweets to avoid GPU memory overflow
- L2-normalised before passing to classifiers
- Same 6 classifiers trained on top of embeddings (identical to Experiment A)
- Embeddings saved as .npy files to roberta_results/ — no need to re-extract on restart

7.3 Cross-Domain Results vs TF-IDF

Model	TF-IDF F1	TF-IDF Rec	RoBERTa F1	RoBERTa Rec	RoBERTa AP	F1 Change
Logistic Regression	0.0119	0.0060	**0.5105**	**0.3615**	0.7515	**+0.4986 (×43)**
SVM (Linear)	0.0146	0.0073	**0.4842**	**0.3339**	0.7545	**+0.4696 (×33)**



📌 ***Recall improved from near-zero (~0.006) to roughly 1 in 3 suicidal tweets correctly identified (~0.36) for LR/SVM. However, KNN showed the most dramatic improvement — rising from dead last (F1: 0.0009) to best overall (F1: 0.6447, Recall: 0.5429). This confirms the TF-IDF failure was a vocabulary mismatch and dimensionality problem, not an inherent impossibility.***

8. Experiment C — MentalBERT Embeddings

8.1 What MentalBERT Does Differently

MentalBERT is a BERT model specifically pre-trained on mental health forum data (Reddit mental health communities, CLPsych shared task datasets). The hypothesis is that domain-specific pre-training on mental health language would give it an advantage over general-purpose RoBERTa for detecting mental health content cross-domain.

- Model: mental/mental-bert-base-uncased (gated on HuggingFace — requires access request)
- Paper: Ji et al. (2022) MentalBERT: Publicly Available Pretrained Language Models for Mental Health Analysis
- Same 6 classifiers, same protocol as Experiments A and B
- Embeddings saved to mentalbert_results/

8.2 Surprising Finding — RoBERTa Outperformed MentalBERT

Model	RoBERTa F1	MentalBERT F1	RoBERTa Rec	MentalBERT Rec	Winner
Logistic Regression	**0.5105**	0.4249	**0.3615**	0.2811	**RoBERTa**
SVM (Linear)	**0.4842**	0.2764	**0.3339**	0.1640	**RoBERTa**

Notably, KNN also achieved the best cross-domain performance within MentalBERT embeddings (F1: 0.6280, Recall: 0.5200), surpassing LR and SVM by a significant margin , consistent with the pattern observed in Experiment B.

8.3 Why RoBERTa Won

- RoBERTa was trained on 160GB of text with a more robust training procedure. MentalBERT's domain-specific corpus is far smaller. Training data size often matters more than domain specificity.Scale advantage:
- MentalBERT was pre-trained on forum text (Reddit) which is longer and more structured. The evaluation tweets are informal short-form Twitter posts, RoBERTa's general web training fits this register better.Register mismatch:
- Both models here are used as frozen feature extractors only. MentalBERT would likely close or reverse this gap with end-to-end fine-tuning on the downstream task.Frozen vs fine-tuned.

9. Final Three-Way Comparison

A unified comparison was produced loading results from all three experiments and comparing all 6 classifiers across all 3 feature types on both in-domain and cross-domain tasks.

9.1 Complete Cross-Domain Results — All Classifiers

Model	TF-IDF F1	TF-IDF Rec	TF-IDF AP	RoBERTa F1	RoBERTa Rec	RoBERTa AP	MentalBERT F1	MentalBERT Rec	MentalBERT AP
Naive Bayes	0.2028	0.1259	0.3894	0.4347	0.2898	0.7419	0.4845	0.3519	0.6801
Random Forest	0.0998	0.0528	0.5989	0.3517	0.2196	0.7151	0.3660	0.2315	0.7081
Decision Tree	0.0964	0.0510	0.3426	0.3974	0.2954	0.4001	0.3703	0.2632	0.3951
SVM (Linear)	0.0146	0.0073	0.4514	0.4842	0.3339	0.7545	0.2764	0.1640	0.6732
Logistic Regression	0.0119	0.0060	0.4448	0.5105	0.3615	0.7515	0.4249	0.2811	0.7264
KNN	0.0009	0.0005	0.3123	**0.6447**	**0.5429**	0.6612	**0.6280**	**0.5200**	0.6453

9.2 Cross-Domain Performance Hierarchy (Corrected)

The corrected hierarchy below reflects actual results from the full comparison table. The previous version erroneously omitted KNN with transformer embeddings, which achieved the highest cross-domain F1 and Recall of any classifier-feature combination:

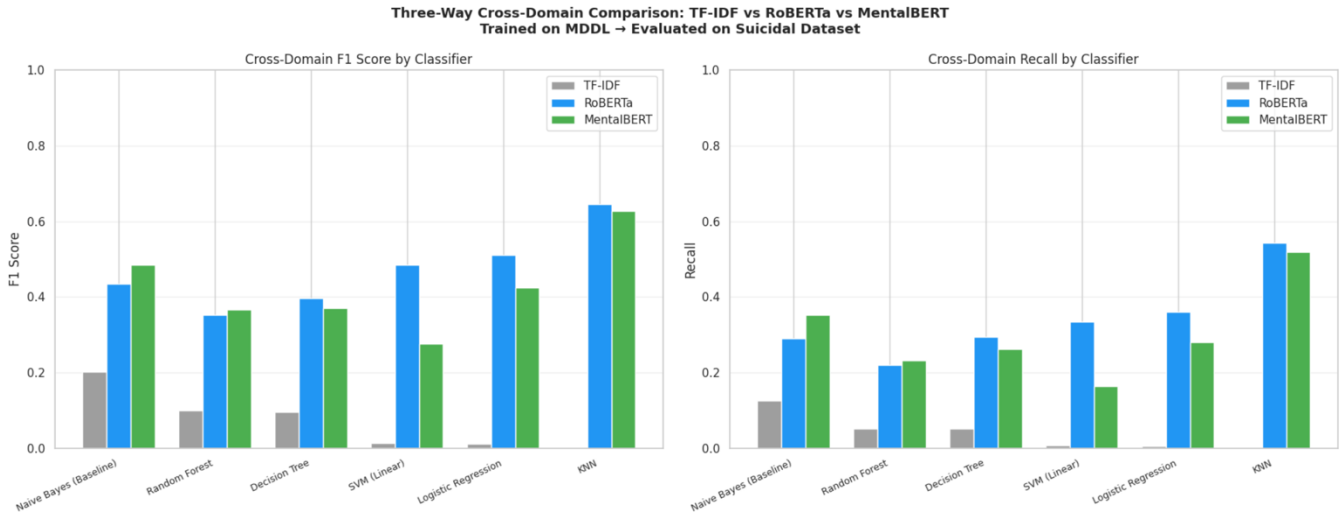
Tier	Method	Best F1	Best Recall	Best AP
3rd	**TF-IDF**	0.2028 (Naive Bayes)	0.1259 (Naive Bayes)	0.5989 (Random Forest)
2nd	**MentalBERT**	0.6280 (KNN)	0.5200 (KNN)	0.7264 (LR)
1st	**RoBERTa**	**0.6447 (KNN)** 🥇	**0.5429 (KNN)** 🥇	0.7545 (SVM)

9.3 The KNN Reversal — Most Striking Finding

KNN's performance trajectory across feature types represents the most dramatic individual finding of the entire study:

Feature Method	KNN F1	KNN Recall	Best Other F1	Best Other Recall	KNN Status
TF-IDF	0.0009	0.0005	0.2028 (NB)	0.1259 (NB)	❌ Dead Last
MentalBERT	**0.6280**	**0.5200**	0.4249 (LR)	0.2811 (LR)	🥇 **Best**
RoBERTa	**0.6447**	**0.5429**	0.5105 (LR)	0.3615 (LR)	🥇 **Best**

This reversal (from absolute worst to absolute best)perfectly illustrates the key lesson of the project: feature representation matters more than classifier choice for cross-domain NLP. With TF-IDF, KNN's core assumption (that nearby points are semantically similar) completely breaks down in high-dimensional sparse space. With dense 768-dimensional transformer embeddings, nearest neighbours are genuinely semantically similar tweets, and KNN exploits this effectively.



10. What This Means for the Final Report

10.1 Introduction / Motivation

Depression and suicidal ideation are related but distinct mental health phenomena. If classifiers trained on depression data could detect suicidal content, it would enable early intervention without requiring labelled suicidal data, which is expensive and ethically sensitive to collect. The results show this transfer is very difficult with traditional methods (TF-IDF) but substantially improved with contextual embeddings (RoBERTa, MentalBERT).

10.2 Key Contributions

- Systematic comparison of TF-IDF vs two transformer models on the same cross-domain task with identical experimental conditions — 6 classifiers each, same split, same metrics
- Vocabulary overlap analysis that quantifies the linguistic gap between domains (40% shared top-50 words) and directly motivates the model hierarchy
- AP Score analysis revealing that model failure at a fixed threshold does not mean model failure at the probability level — Random Forest has the best TF-IDF AP despite one of the worst F1 scores
- Empirical evidence that general-purpose transformers (RoBERTa) outperform domain-specific ones (MentalBERT) in feature-extraction mode due to training scale
- Discovery of the KNN reversal: KNN goes from worst performer (F1: 0.0009 with TF-IDF) to best performer (F1: 0.6447 with RoBERTa), demonstrating that feature representation quality is the dominant factor in cross-domain transfer

10.3 Limitations to Discuss

- All transformer models used as frozen feature extractors only, fine-tuning would likely improve results significantly
- Best cross-domain Recall is ~0.54 (KNN + RoBERTa), meaning roughly 1 in 2 suicidal tweets are still missed, not deployment-ready
- KNN's best cross-domain performance comes with a trade-off: higher computational cost at inference time compared to linear models